

Time-series and statistical data

October 2, 2025

Contents

1	Ambient condition random process	1
2	Wind Farm Models	2
3	Accumulated Metric	2
4	Expected value for integral metrics	3
4.1	Simple Time-Integration	3
4.2	Discounted Time-Integration	4

1 Ambient condition random process

Let

$$\mathbf{A}_t = \begin{pmatrix} A_{1,t} \\ \vdots \\ A_{S_a,t} \end{pmatrix} \quad (1)$$

denote an S_a -dimensional vector-valued random process with time index t , where each $a_s, s \in \{1, \dots, S\}$ represents an ambient condition according to Confluence: Ambient Conditions.

We assume that $\mathbf{A}_t$ is stationary, such that its joint probability density function

$$f(\mathbf{a}) := f(a_0, \dots, a_{S-1}) \quad (2)$$

does not change over time.

In practice, instead of a continuous pdf $f(\mathbf{a})$, we are often given a discrete probability mass function (PMF)

$$\rho_i := \Pr(\mathbf{A}_t = \mathbf{a}_i), \quad i \in 1, \dots, I, \quad (3)$$

$$\text{with } \sum_{i=1}^I \rho_i = 1. \quad (4)$$

2 Wind Farm Models

A wind farm model (WFM) is represented by a vector-valued function $\mu : \mathbb{R}^{S_c} \times \mathbb{R}^{S_a} \rightarrow \mathbb{R}^{S_o}$

$$\mathbf{y} = \mu(\mathbf{u}, \mathbf{a}), \quad (5)$$

where $\mathbf{u} \in \mathbb{R}^{S_c}$, $\mathbf{a} \in \mathbb{R}^{S_a}$ and $\mathbf{y} \in \mathbb{R}^{S_o}$ denote the control input, ambient condition and model output vector, respectively.

We further define a control policy $\pi : \mathbb{R}^{S_a} \rightarrow \mathbb{R}^{S_c}$, which assigns a control input

$$\mathbf{u} = \pi(\mathbf{a}) \quad (6)$$

to each ambient condition. The considered control policies are *static* in the sense that they map exactly one control input to each ambient condition at all times.

If the statistics of $\mathbf{A}_t$ are given in the form of a discrete PMF, the static control policy is denoted

$$\mathbf{u} := (\mathbf{u}_1,) \quad (7)$$

For a given control policy π , we can define a parametrized version of the WFM (PWFM) that is a function only of the ambient condition,

$$\mu^\pi : \mathbb{R}^{S_a} \rightarrow \mathbb{R}^{S_o}, \quad (8)$$

such that

$$\mathbf{y} = \mu^\pi(\mathbf{a}) := \mu(\pi(\mathbf{a}), \mathbf{a}). \quad (9)$$

3 Accumulated Metric

For a given time interval $t \in [0, T]$ and model output $\mathbf{y}(t)$ we define a vector of integral-metrics

$$\mathbf{m}(T) = \int_0^T \mathbf{y}(t) \, dt. \quad (10)$$

For given ambient condition data and $\mathbf{a}(t), t \in [0, T]$ and control policy π , the integral metric reads

$$\mathbf{m}(T) = \int_0^T \mu(\mathbf{a}(t)) \, dt, \quad (11)$$

where the superscript in μ^π has been omitted for legibility.

Remark 1 *The concept of integral-metrics can be extended to more general accumulated metrics,*

$$\mathbf{m}^\phi(T) = \int_0^T \phi(\mathbf{y}(t), t) \, dt, \quad (12)$$

where ϕ denotes a vector-valued function of the model outputs and time. This can be used for example to consider discounting rates over time.

4 Expected value for integral metrics

4.1 Simple Time-Integration

We are interested in the expected value of an integral metric $\mathbb{E}[\mathbf{m}(T)]$, where the expectation is taken over $f(\mathbf{a})$. We have

$$\mathbb{E}[\mathbf{m}(T)] = \mathbb{E}\left[\int_0^T \mu(\mathbf{a}(t)) \, dt\right] \quad (13)$$

$$= \int_0^T \mathbb{E}[\mu(\mathbf{a}(t))] \, dt \quad (14)$$

due to fundamental rules of integration and linearity of the expectation value. Due to stationarity of the ambient conditions, we have

$$\mathbb{E}[\mu(\mathbf{a}(t))] = \mathbb{E}[\mu(\mathbf{a})], \quad (15)$$

which is a constant (in time t), and the expected value is computed as

$$\mathbb{E}[\mathbf{m}(T)] = T \, \mathbb{E}[\mu(\mathbf{a})] \quad (16)$$

$$= T \int_{S(\mathbf{a})} \mu(\mathbf{a}) \, f(\mathbf{a}) \, d\mathbf{a}. \quad (17)$$

If, instead of a continuous pdf $f(\mathbf{a})$, a discrete point mass distribution

$$\rho_i := \Pr(\mathbf{A} = \mathbf{a}_i), \quad i \in 1, \dots, I, \quad (18)$$

$$\text{with } \sum_{i=1}^I \rho_i = 1 \quad (19)$$

is given, then the expectation is computed as

$$\mathbb{E}[\mathbf{m}(T)] = T \sum_{i=1}^I \mu^\pi(\mathbf{a}_i) \rho_i. \quad (20)$$

4.2 Discounted Time-Integration

We further consider the case where a discount factor $0 < \gamma \leq 1$ is applied over time,

$$\phi(\mathbf{y}, t) = \gamma^t \mathbf{y}. \quad (21)$$

The expectation of the accumulated metric then reads according to (14)

$$\mathbb{E} [\mathbf{m}^\phi(T)] = \int_0^T \mathbb{E} [\phi(\mu(\mathbf{a}(t)), t)] \, dt \quad (22)$$

$$= \int_0^T \mathbb{E} [\gamma^t \mu(\mathbf{a}(t))] \, dt \quad (23)$$

$$= \mathbb{E} [\mu(\mathbf{a})] \int_0^T \gamma^t \, dt \quad (24)$$

$$= \mathbb{E} [\mu(\mathbf{a})] \frac{\gamma^T - 1}{\log(\gamma)} \quad (25)$$

If instead the discount is defined in discrete time t ,

$$\mathbf{m}^\phi(T) = \sum_{t=0}^{T-1} \gamma^t \mu(\mathbf{a}(t)) \quad (26)$$

we get

$$\mathbb{E} [\mathbf{m}^\phi(T)] = \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t \mu(\mathbf{a}(t)) \right] \quad (27)$$

$$= \mathbb{E} [\mu(\mathbf{a})] \sum_{t=0}^{T-1} \gamma^t \quad (28)$$

$$= \mathbb{E} [\mu(\mathbf{a})] \frac{1 - \gamma^T}{1 - \gamma}. \quad (29)$$

The discount factor could be derived from a yearly discount rate, for example, in which case the model output $\mu(\mathbf{a})$ needs to correspond to a yearly rate.

Note that the yearly discount factor is usually computed from a yearly discount rate r ,

$$\gamma = \frac{1}{1 + r}. \quad (30)$$

It is also important to note that the effect of the discount is applied as an external factor to the expectation value in both the discrete and continuous case.